

Cybersecurity Incident Report

Section 1: Identify the type of attack that may have caused this network interruption

One potential explanation for the website's connection timeout error message is a DoS attack.

The logs show that the web server stopped responding after it is overloaded with SYN packet requests.

This event could be a type of DoS attack called SYN flooding.

Section 2: Explain how the attack is causing the website to malfunction

When website visitors try to establish a connection with the web server, a three-way handshake occurs using the TCP protocol. The handshake consists of three steps:

1. The client sends a SYN packet to the server, requesting to connect;
2. The web server responds with a SYN/ACK packet to accept the connection request;
3. The client sends back an ACK packet to the server. The connection is established.

When a malicious actor sends a large number of SYN packets all at once, it consumes the web server's available connection resources and fills its TCP connection backlog. As a result, legitimate clients may be unable to establish new TCP connections and eventually experience connection timeouts.

The logs indicate that the web server has become overwhelmed and is unable to treat the visitor's SYN requests. The server is unable to open a new connection to visitors who receive a connection timeout message.